📊 Session 6 – Data Model Tables (Star Schema)

1️⃣ Fact_WorkOrders Table

Column Name	Data Type	Description	Key Type
work_order_id	Text	Unique identifier for each work order	Primary Key
asset_id	Text	Asset related to the work order	Foreign Key
technician_id	Text	Assigned technician	Foreign Key
priority	Text (P1–P4)	Urgency level for SLA selection	-
status	Text (Open/In Progress/Closed)	Current work order status	-
created_at	DateTime	Timestamp when work order was created	-
started_at	DateTime	Timestamp when repair started	-
closed_at	DateTime	Timestamp when work order was closed	-
downtime_minutes	Whole Number	Total downtime caused by issue	-
sla_target_minutes	Whole Number	SLA target based on priority	-

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2️⃣ Dim_Assets Table

Column Name	Data Type	Description	Key Type
asset_id	Text	Unique asset identifier	Primary Key
asset_name	Text	Name of the asset	-
location	Text	Asset location/site	-
criticality	Text (High/Medium/Low)	Business impact level	-
asset_type	Text	Category of asset	-

3️⃣ Dim_Technicians Table

Column Name	Data Type	Description	Key Type
technician_id	Text	Unique technician ID	Primary Key
technician_name	Text	Full name	-
team	Text	Team assignment	-
shift	Text	Working shift	-

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4 Dim_SLA Table

Column Name	Data Type	Description	Key Type
priority	Text	Priority level (P1–P4)	Primary Key
sla_target_minutes	Whole Number	SLA target duration in minutes	-
description	Text	SLA rule explanation	-

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5 Dim_Date Table

Column Name	Data Type	Description	Key Type
date	Date	Calendar date	Primary Key
day	Whole Number	Day of month	-
month	Text	Month name	-
month_number	Whole Number	Month number	-
quarter	Text	Quarter (Q1–Q4)	-
year	Whole Number	Year	-
week_number	Whole Number	Week of year	-

🔗 Relationships (Model Rules)

From Table	Column	To Table	Column	Relationship
Dim_Assets	asset_id	Fact_WorkOrders	asset_id	One-to-Many
Dim_Technicians	technician_id	Fact_WorkOrders	technician_id	One-to-Many
Dim_SLA	priority	Fact_WorkOrders	priority	One-to-Many

From Table	Column	To Table	Column	Relationship
Dim_Date	date	Fact_WorkOrders	created_at (date)	One-to-Many

📌 **Required KPI Measures Table**

KPI Name	Formula Logic	Based On Fields
Total Work Orders	COUNTROWS(Fact_WorkOrders)	work_order_id
Closed Work Orders	Status = "Closed"	status
Open Work Orders (Backlog)	Status ≠ "Closed"	status
SLA Compliance %	On-time Closed / Total Closed	created_at, closed_at, sla_target_minutes
SLA Breaches	Closed – On-time Closed	created_at, closed_at
Total Downtime	SUM(downtime_minutes)	downtime_minutes
MTTR (hours)	Avg(closed_at – started_at)	started_at, closed_at